



Fig. 3: Nautilus speaker of Bower & Wilkins



Fig. 4: Two-way spiral speaker

three metres length, which is impractical. But curled up horn shape with a height of 1.2 metres, like a nautilus, meets the demand. Thus, there is a great saving in space.

One key advantage of a TL speaker is its ability to conduct the back waves behind the driver more effectively away from it, reducing the chances for reflected energy permeating back through the diaphragm out of phase with the primary signal. The second



Fig. 5: Spirula speaker made by Deeptime, Prague



Fig. 6: Vivid transmission line speaker

benefit is that the resulting audio time is coherent, that is, in phase. A TL speaker can produce the full range of sound without using a subwoofer.

Makers of nautilus shaped speakers

Well-designed nautilus shaped speakers are legendary in their performance. These can be manufactured by moulding or by hot press, but most conveniently by 3D printing. 3D printing is a way of creating three-dimensional solid objects by building up the object layer by layer using a 3D printer and a computer.

Nautilus speakers are made among others by Bowers & Wilkins, a British company, who are pioneers in making the iconic shaped speakers shown in Fig. 3. It is a 4-way speaker system, where the woofer is housed in the logarithmic spiral and three other mid-range speakers and tweeters are placed in individual tapered tube housings, independent of the nautilus shaped woofer enclosure, at the top. It is quite big with 1210mm height, 430mm width, 1105mm depth, and weighing about 87kg. It costs a fortune, about \$70,000 a pair. The cost is so high because of the expensive materials used, costly machines and tools required to make them, and every part is hand-crafted to achieve better sound quality and reliability, which is further augmented by marketing and branding expenses.

Another variant of two-way spiral speakers is shown in Fig. 4.

A small company called Deeptime in Prague, a Czech design studio, also makes the logarithmic spiral speakers. They call it 'Spirula,' inspired by the geometry of deep-sea squid named Spirula. It is made by 3D printing using sand mixed with hardener (Fig. 5). It has excellent damping property due to use of absorbent fibre. It is a bookshelf type speaker costing around \$1000 with a frequency range of 75Hz to 20kHz. It uses only one 7.5cm (3-inch) full range driver.

Vivid Audio, UK also makes similar transmission line speakers with exotic curves, as shown in Fig. 6. It is a four-to-six driver, three-way, floor-standing speaker with its internal volume of 90, 45, or 25 litres. It embodies the same engineering philosophy as the nautilus. It costs about \$26,000 for its top-line model. The cabinets are constructed out of a glass-reinforced Soric-cored sandwich composite. Soric is a honeycomb shell fabric that is designed for laminates. Three layers of the fabric are placed into an airtight mould, the air is pumped out and the bonding agent is pumped in. The result is a light and rigid panel.

Many audio enthusiasts make nautilus shaped speakers at home such as the one in Fig. 7. It has been reported that it took 400 hours of work and 112 wooden discs, every one different in inside and outside diameter,